

PROFILE

Software Engineer with 2+ years of experience in Java, Spring Boot, and Python, building scalable backend systems, microservices, and REST APIs. Experience with distributed systems, SQL/NoSQL databases, cloud platforms, and containerization, along with hands-on experience in RAG and LLM-based applications.

TECHNICAL SKILLS

- Languages: Java, Python, SQL, C#
- Frameworks: Spring Boot, FastAPI, React, LangChain
- Backend: REST APIs, Hibernate, JPA
- Databases: PostgreSQL, MySQL, MongoDB
- Cloud & Infrastructure: Azure, Docker, Linux
- Tools: Git, IntelliJ, Maven, CI/CD
- Concepts: System Design, Distributed Systems, Microservices Architecture, Data Structures & Algorithms, Dependency Injection

EDUCATION

- Bauhaus-Universität

Master of Science in Digital Engineering (Grade: 90%)

• Relevant Coursework: Machine Learning, Computer Vision, Simulation Methods, Stochastic Systems, Optimization, Data Structures & Algorithms, Object-Oriented Programming, Software Engineering

Chennai, India

2023 – 2026
- Anna University

Bachelor of Engineering in Civil Engineering (Grade: 82%)

• Co-author, *Int. J. Steel Struct.* (2024), DOI: 10.1007/s13296-024-00816-9

Weimar, Germany

2017 – 2021

PROFESSIONAL EXPERIENCE

- FORVIA Hella GmbH & Co. KGaA

Software Developer Intern / Master's Thesis

• Designed and built a distributed Python workflow engine to orchestrate automotive simulation jobs across HPC clusters.

• Developed modular backend components for job scheduling, execution monitoring, and result aggregation.

• Automated end-to-end simulation pipelines, eliminating 90% of manual processes and reducing overhead

• Integrated external APIs and data pipelines to connect simulation infrastructure with downstream systems.

• Refactored legacy simulation scripts into structured backend services, improving scalability and reducing technical debt.

10/2024 – 09/2025

Lippstadt, Germany
- Bauhaus-Universität Weimar

Student Assistant

• Enhanced a Retrieval-Augmented Generation (RAG) system by integrating external knowledge sources and improving retrieval quality.

• Supported evaluation and teaching activities at the Chair of Stochastics and Optimization.

04/2024 – 09/2024

Weimar, Germany
- Tata Consultancy Services

Assistant System Engineer

• Developed and maintained Java Spring Boot microservices for a US-based financial services client, supporting high-volume transaction processing across distributed systems.

• Implemented automated reconciliation pipelines to validate financial transactions against external records, identifying discrepancies and reducing manual exception handling.

• Built backend APIs and batch processing workflows to handle large-scale financial data, ensuring accuracy and reliability across end-of-day settlement cycles.

04/2021 – 02/2023

Chennai, India

PROJECTS

- RAG Chatbot Backend Application | *LangChain, FastAPI, Python, ChromaDB, LLM*

• Built a Retrieval-Augmented Generation (RAG) system enabling efficient querying of large document datasets

• Implemented backend APIs for document ingestion, vector indexing, and LLM-based response generation.

• Designed conversational workflows with chat history storage using SQLite and support for follow-up queries
- Order Management System | *Spring Boot, Java, PostgreSQL*

• Developed a backend system to manage users, products, and orders using REST APIs

• Designed relational data models with entities such as Order, Product, and User using Spring Data JPA

• Implemented order workflows including item management, total price calculation, and status tracking

COMPETITIONS

- Participated in TREC 2024 RAG Track, organized by National Institute of Standards and Technology (NIST), USA and University of Waterloo.

LANGUAGES

English (C1), German (B1/B2), Tamil (Native)